

1. Library Book Management using Classes & Encapsulation

```
import java.util.Scanner;

class Book {
    private int id;
    private String title;
    private boolean issued;

    Book(int id, String title) {
        this.id = id;
        this.title = title;
        issued = false; }

    int getId() {
        return id; }

    void issue() {
        if (!issued) {
            issued = true;
            System.out.println("Book Issued");
        } else
            System.out.println("Already Issued");
    }

    void returnBook() {
        if (issued) {
            issued = false;
            System.out.println("Book Returned");
        } else
            System.out.println("Book Not Issued"); }

    void display() {
        System.out.println(id + " " + title + " " + (issued ? "Issued" : "Available"));
    }
}

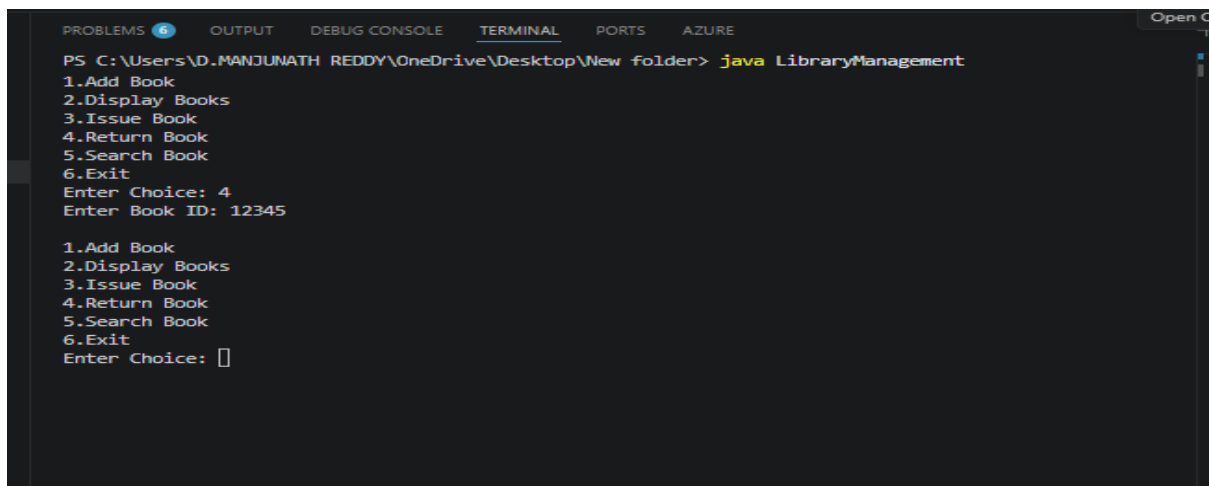
public class LibraryManagement {
    public static void main(String[] args) {
```



```

        System.out.print("Book ID: ");
        id = sc.nextInt();
        for (int i = 0; i < n; i++)
            if (b[i].getId() == id)
                b[i].returnBook();
        break;
case 5:
    System.out.print("Book ID: ");
    id = sc.nextInt();
    for (int i = 0; i < n; i++)
        if (b[i].getId() == id)
            b[i].display();
    break;
case 6:
    System.out.println("Thank You");
    break;
default:
    System.out.println("Invalid Choice");
}} while (ch != 6);
sc.close();}}

```



The screenshot shows a terminal window with the following content:

```

PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java LibraryManagement
1.Add Book
2.Display Books
3.Issue Book
4.Return Book
5.Search Book
6.Exit
Enter Choice: 4
Enter Book ID: 12345

1.Add Book
2.Display Books
3.Issue Book
4.Return Book
5.Search Book
6.Exit
Enter Choice: 

```

2. Employee Payroll System using Inheritance

```
class Employee {
    int id;
    String name;
    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
    double calculateSalary() {
        return 0;
    }
    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Salary: $" + calculateSalary());
    }
}

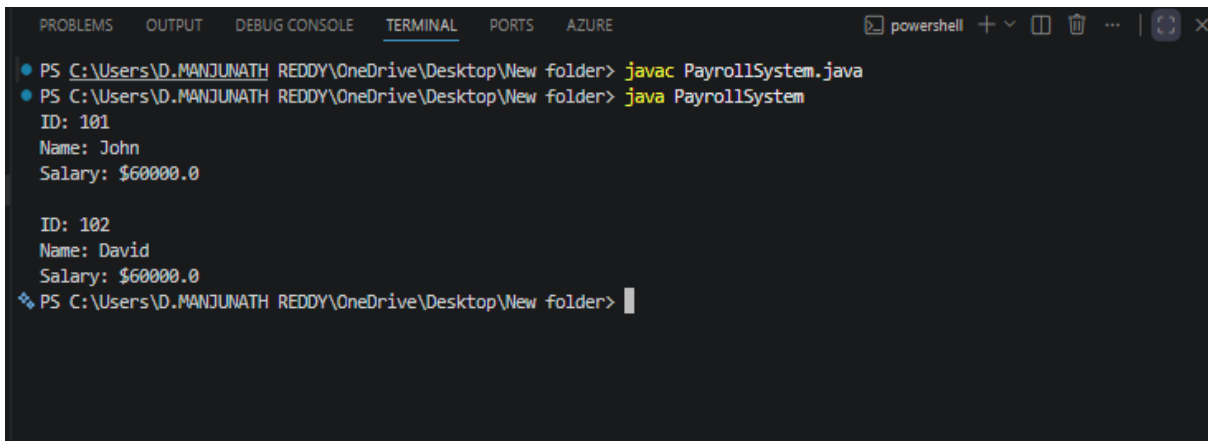
class PermanentEmployee extends Employee {
    double basic;
    PermanentEmployee(int id, String name, double basic) {
        super(id, name);
        this.basic = basic;
    }
    @Override
    double calculateSalary() {
        return basic + basic * 0.20;
    }
}
```

```

class ContractEmployee extends Employee {
    int hours;
    double rate;
    ContractEmployee(int id, String name, int hours, double rate) {
        super(id, name);
        this.hours = hours;
        this.rate = rate;
    }
    @Override
    double calculateSalary() {
        return hours * rate;
    }
}

public class PayrollSystem {
    public static void main(String[] args) {
        Employee e1 = new PermanentEmployee(101, "John", 50000);
        Employee e2 = new ContractEmployee(102, "David", 120, 500);
        e1.display();
        System.out.println();
        e2.display();
    }
}

```



```

PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> javac PayrollSystem.java
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java PayrollSystem
ID: 101
Name: John
Salary: $60000.0

ID: 102
Name: David
Salary: $60000.0
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder>

```

3. Vehicle Rental System using Runtime Polymorphism

```
class Vehicle {
    String name;

    Vehicle(String name) {
        this.name = name;
    }

    double calculateRent(int days) {
        return 0;
    }

    void bill(int days) {
        System.out.println(name);
        System.out.println("Days: " + days);
        System.out.println("Rent: $" + calculateRent(days));
        System.out.println();
    }
}

class Car extends Vehicle {
    Car() {
        super("Car");
    }

    @Override
    double calculateRent(int days) {
        return days * 1000;
    }
}

class Bike extends Vehicle {

    Bike() {
        super("Bike");
    }

    @Override
    double calculateRent(int days) {
```

```

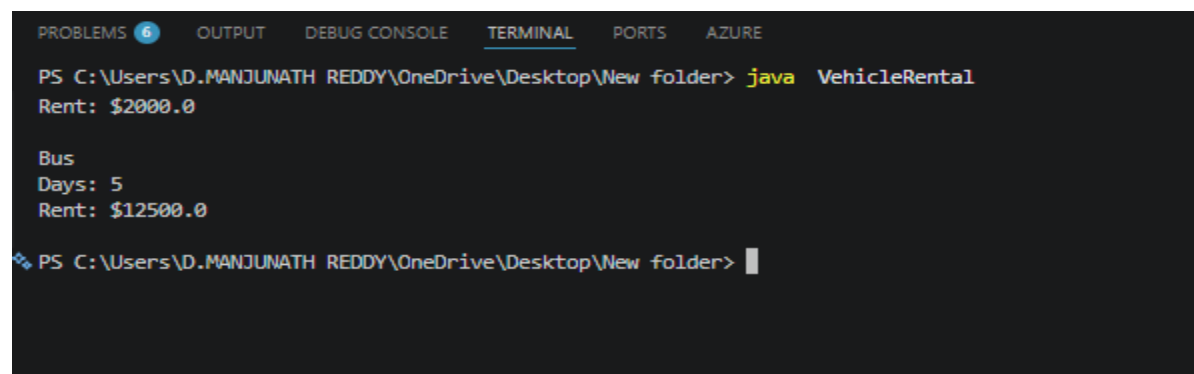
        return days * 400;
    }
}

class Bus extends Vehicle {
    Bus() {
        super("Bus");
    }

    @Override
    double calculateRent(int days) {
        return days * 2500;
    }
}

public class VehicleRental {
    public static void main(String[] args) {
        Vehicle[] vehicles = {
            new Car(),
            new Bike(),
            new Bus()
        };
        int days = 5;
        for (Vehicle v : vehicles) {
            v.bill(days);
        }
    }
}

```



```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java VehicleRental
Rent: $2000.0

Bus
Days: 5
Rent: $12500.0
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder>

```

4. Bank Account System using Data Hiding

```
import java.util.Scanner;

class BankAccount {
    private int accNo;
    private String name;
    private double balance;

    BankAccount(int accNo, String name, double balance) {
        this.accNo = accNo;
        this.name = name;
        this.balance = balance;
    }

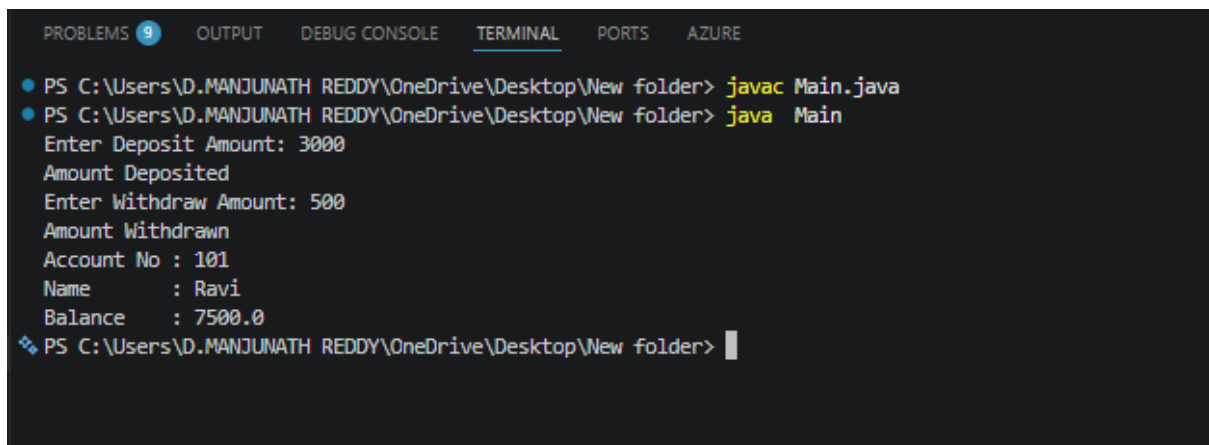
    void deposit(double amount) {
        balance += amount;
        System.out.println("Amount Deposited");
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println("Amount Withdrawn");
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    void display() {
        System.out.println("Account No : " + accNo);
        System.out.println("Name      : " + name);
        System.out.println("Balance   : " + balance);
    }
}
```



```
}  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        BankAccount b = new BankAccount(101, "Ravi", 5000);  
        System.out.print("Enter Deposit Amount: ");  
        b.deposit(sc.nextDouble());  
        System.out.print("Enter Withdraw Amount: ");  
        b.withdraw(sc.nextDouble());  
        b.display();  
        sc.close();  
    }  
}
```



The screenshot shows a terminal window with the following content:

```
PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE  
● PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> javac Main.java  
● PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java Main  
Enter Deposit Amount: 3000  
Amount Deposited  
Enter Withdraw Amount: 500  
Amount Withdrawn  
Account No : 101  
Name      : Ravi  
Balance   : 7500.0  
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> |
```

5. University Admission System using Method Overloading

```
class Admission {  
    void calculateFee() {  
        System.out.println("UG Fee : 50000");  
    }  
    void calculateFee(int pg) {  
        System.out.println("PG Fee : 70000");  
    }  
    void calculateFee(String scholarship) {  
        System.out.println("Scholarship Fee : 20000");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Admission a = new Admission();  
        a.calculateFee();  
        a.calculateFee(1);  
        a.calculateFee("Yes");  
    }  
}
```



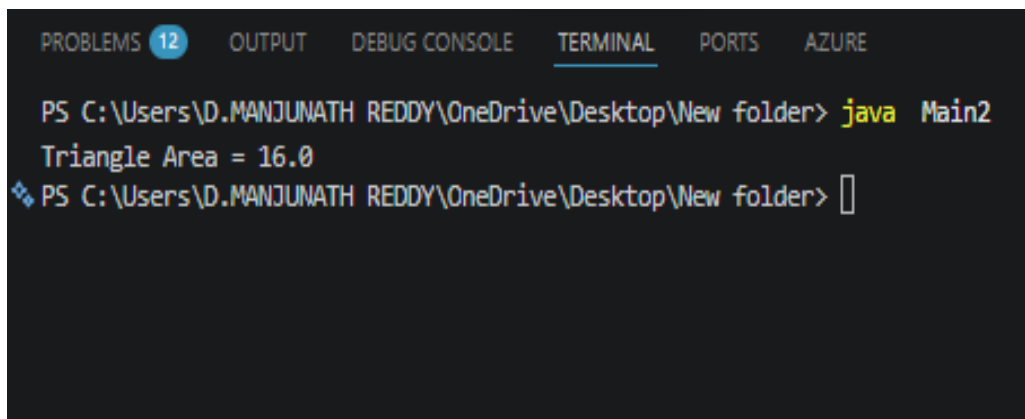
The screenshot shows a Java IDE with a terminal window. The terminal displays the output of the program, which is the result of running the `java Main1` command. The output shows the fees for different admission types: UG Fee (50000), PG Fee (70000), and Scholarship Fee (20000). The terminal also shows the command prompt and the command used to run the program.

```
PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java Main1  
PG Fee : 70000  
Scholarship Fee : 20000  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder>
```

6. Shape Area Calculator using Abstract Class & Polymorphism

```
abstract class Shape {  
    abstract void area();  
}  
  
class Circle extends Shape {  
    double r = 5;  
    void area() {  
        System.out.println("Circle Area = " + (3.14 * r * r));  
    }  
}  
  
class Rectangle extends Shape {  
    int l = 4, b = 6;  
    void area() {  
        System.out.println("Rectangle Area = " + (l * b));  
    }  
}  
  
class Triangle extends Shape {  
    int base = 4, height = 8;  
    void area() {  
        System.out.println("Triangle Area = " + (0.5 * base * height));  
    }  
}
```

```
public class Main2 {  
    public static void main(String[] args) {  
  
        Shape[] s = {  
            new Circle(),  
            new Rectangle(),  
            new Triangle()  
        };  
  
        for (Shape x : s)  
            x.area();  
    }  
}
```

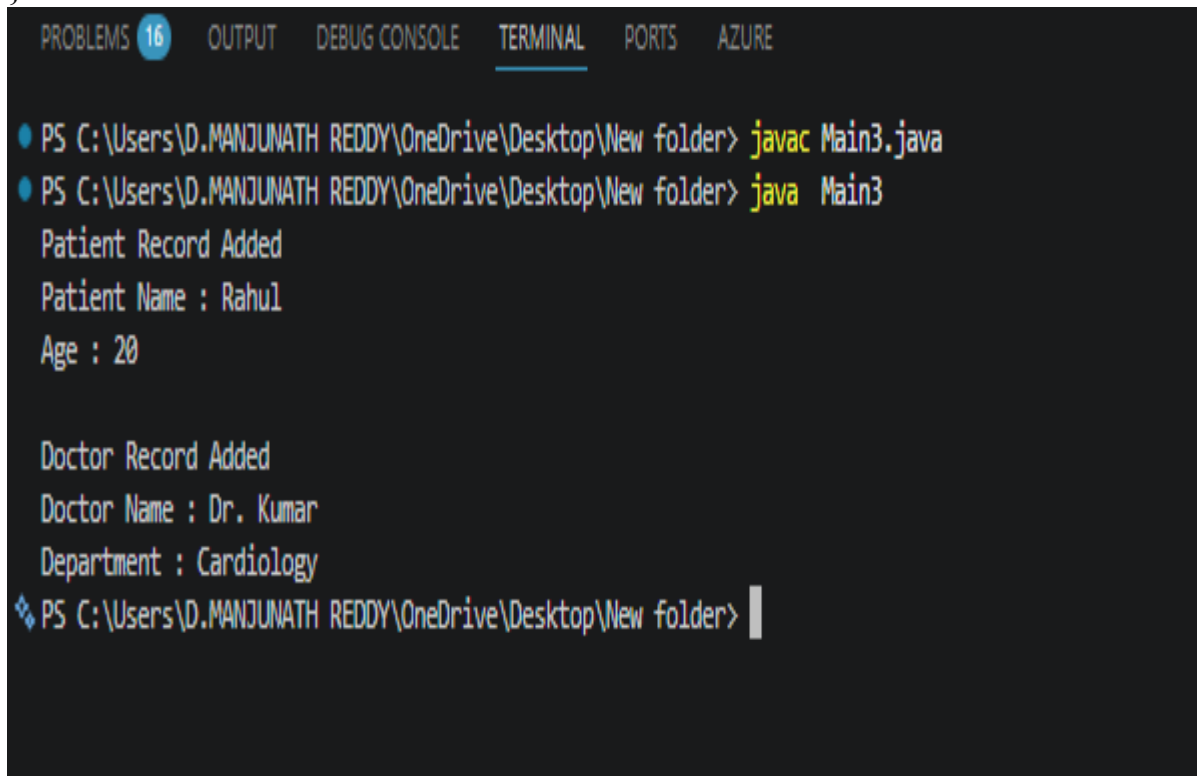
A screenshot of a Java IDE's terminal window. The terminal has tabs for PROBLEMS (12), OUTPUT, DEBUG CONSOLE, TERMINAL (selected), PORTS, and AZURE. The command prompt shows the execution of 'java Main2' in the directory 'C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder'. The output is 'Triangle Area = 16.0'.

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java Main2  
Triangle Area = 16.0  
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> 
```

7. Hospital Management System using Interfaces

```
interface MedicalRecord {  
    void addRecord();  
    void displayRecord();  
}  
  
class Patient implements MedicalRecord {  
    String name = "Rahul";  
    int age = 20;  
    public void addRecord() {  
        System.out.println("Patient Record Added");  
    }  
    public void displayRecord() {  
        System.out.println("Patient Name : " + name);  
        System.out.println("Age : " + age);  
    }  
}  
  
class Doctor implements MedicalRecord {  
    String name = "Dr. Kumar";  
    String dept = "Cardiology";  
  
    public void addRecord() {  
        System.out.println("Doctor Record Added");  
    }  
  
    public void displayRecord() {  
        System.out.println("Doctor Name : " + name);  
        System.out.println("Department : " + dept);  
    }  
}
```

```
    }  
}  
public class Main {  
    public static void main(String[] args) {  
        MedicalRecord p = new Patient();  
        MedicalRecord d = new Doctor();  
  
        p.addRecord();  
        p.displayRecord();  
  
        System.out.println();  
  
        d.addRecord();  
        d.displayRecord();  
    }  
}
```



```
PROBLEMS 16 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE  
● PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> javac Main3.java  
● PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java Main3  
Patient Record Added  
Patient Name : Rahul  
Age : 20  
  
Doctor Record Added  
Doctor Name : Dr. Kumar  
Department : Cardiology  
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> |
```

8. Online Shopping Order System using Packages

```
class Product {  
    String name;  
    double price;  
    Product(String name, double price) {  
        this.name = name;  
        this.price = price;  
    }  
}  
  
class Order {  
    void bill(Product p, int qty) {  
        double total = p.price * qty;  
        System.out.println("Product : " + p.name);  
        System.out.println("Price   : " + p.price);  
        System.out.println("Quantity : " + qty);  
        System.out.println("Total   : " + total);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Product p = new Product("Laptop", 50000);  
        Order o = new Order();  
        o.bill(p, 2);  
    }  
}
```



```
Main.java x  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL  
PS C:\Java\Prog8> javac Main.java  
PS C:\Java\Prog8> java Main  
  
Product : Laptop  
Price : 50000.0  
Quantity : 2  
Total : 100000.0  
PS C:\Java\Prog8>   
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Java
```

9. Student Result Analyzer using Arrays of Objects

```
class Student {  
    int roll, String name, int m1, m2, m3;  
    Student(int r, String n, int a, int b, int c) {  
        roll = r;  
        name = n;  
        m1 = a;  
        m2 = b;  
        m3 = c;  
    }  
    int total() {  
        return m1 + m2 + m3;  
    }  
    double average() {  
        return total() / 3.0;  
    }  
    String grade() {  
        if (average() >= 90)  
            return "A";  
        else if (average() >= 75)  
            return "B";  
        else if (average() >= 50)  
            return "C";  
        else  
            return "Fail";  
    }  
    void display() {
```



```

        System.out.println(roll + " " + name);
        System.out.println("Total = " + total());
        System.out.println("Average = " + average());
        System.out.println("Grade = " + grade());
    }
}

public class Main {

    public static void main(String[] args) {

        Student s[] = {
            new Student(1, "Rahul", 90, 80, 85),
            new Student(2, "Asha", 95, 92, 91),
            new Student(3, "Kiran", 70, 75, 80)
        };

        int max = 0;

        for (int i = 0; i < s.length; i++) {

            s[i].display();

            System.out.println();

            if (s[i].total() > s[max].total())

                max = i;

        } System.out.println("Class Topper : " + s[max].name);

    }
}

```

```

PROBLEMS 16 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> javac Main4.java
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java Main4
1 Rahul
Total = 255
Average = 85.0
Grade = B

2 Asha
Total = 278
Average = 92.66666666666667
Grade = A

3 Kiran
Total = 225
Average = 75.0
Grade = B

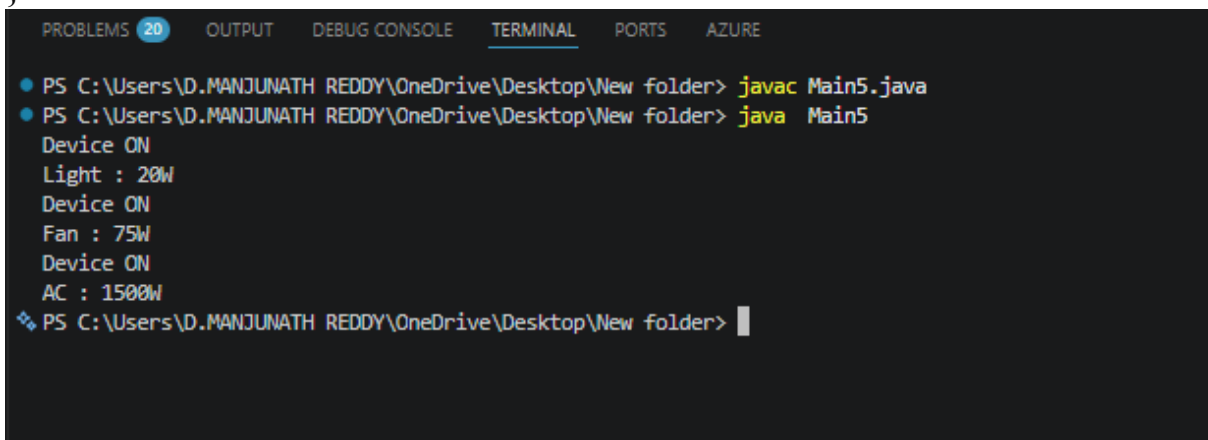
Class Topper : Asha
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder>

```

10. Smart Home Device Controller using Hierarchical Inheritance

```
class Device {  
    void on() {  
        System.out.println("Device ON");  
    }  
  
    void off() {  
        System.out.println("Device OFF");  
    }  
  
    void power() {  
        System.out.println("Power Consumption");  
    }  
}  
  
class Light extends Device {  
    void power() {  
        System.out.println("Light : 20W");  
    }  
}  
  
class Fan extends Device {  
    void power() {  
        System.out.println("Fan : 75W");  
    }  
}
```

```
class AirConditioner extends Device {  
    void power() {  
        System.out.println("AC : 1500W");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Device d;  
        d = new Light();  
        d.on();  
        d.power();  
        d = new Fan();  
        d.on();  
        d.power();  
        d = new AirConditioner();  
        d.on();  
        d.power();  
    }  
}
```



```
PROBLEMS 20 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE  
● PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> javac Main5.java  
● PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder> java Main5  
Device ON  
Light : 20W  
Device ON  
Fan : 75W  
Device ON  
AC : 1500W  
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\New folder>
```